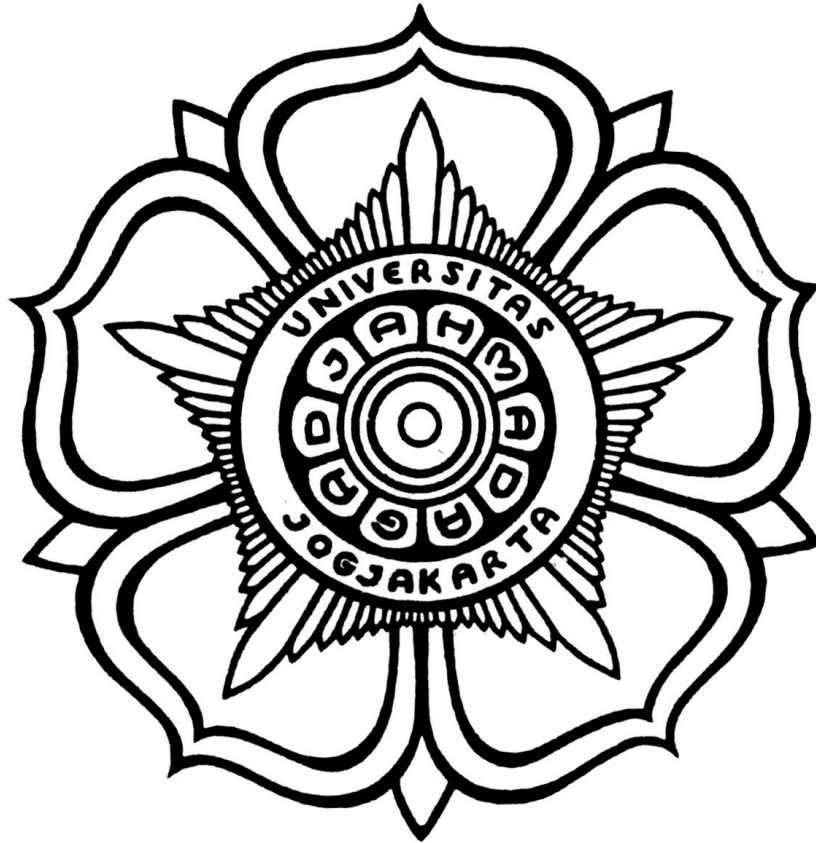


**CAMPUS EVENT MANAGEMENT SYSTEM
SIMAAUGM**



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Date	: 28/12/2025

I. INTRODUCTION

A. Background and Problem

Universitas Gadjah Mada frequently organizes a wide range of student events such as seminars, workshops, and competitions. However, the management of these events — from event creation and venue assignment to participant registration — is often carried out using separate spreadsheets or online forms. This fragmented approach leads to redundant data, inconsistent records, and difficulty in tracking attendance and event outcomes.

SIMAA UGM (Sistem Manajemen Acara UGM) is designed to address these issues by providing a centralized, structured, and efficient system for managing campus events. The system enables organizers to create and manage events, allows students to register for participation, and records venue details systematically, ensuring consistency and reliability of data within a unified database.

B. Objectives

The main objectives of SIMAA UGM are to:

- i. To design a simple yet structured database system for storing event and registration data.
- ii. To support essential CRUD operations for users, events, and registrations.
- iii. Maintain data integrity by linking event, venue, and registration records accurately.
- iv. Provide a scalable foundation for future enhancements such as feedback collection and reporting features.

C. Users and Roles

In this version, the system distinguishes between Students and Organizers as separate entities.

- Organizer: Responsible for creating and managing events, including setting details such as date, location, and capacity..
- Student: Views and registers for available events.

D. Scope

The initial version of SIMAA UGM focuses on three core functionalities:

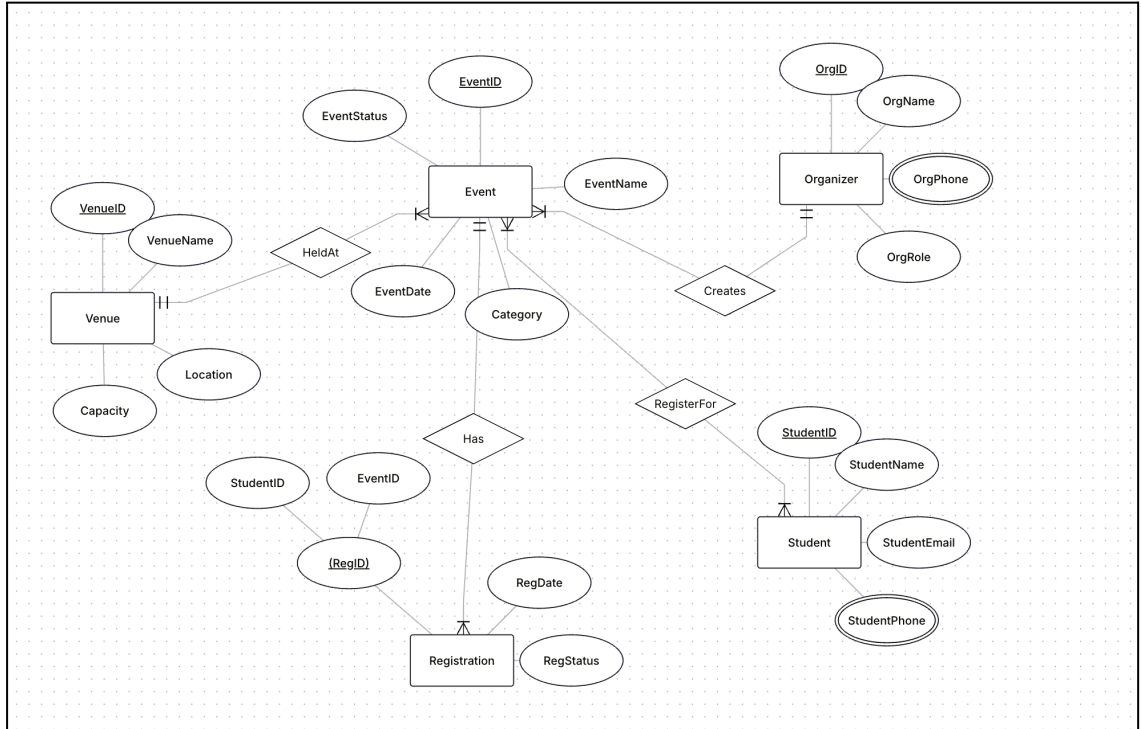
- i. Managing organizer and student information.
- ii. Creating and managing event details, including the assigned venue and event schedule.
- iii. Handling participant registration for events and maintaining registration status records.

Out of scope for this version: feedback management, payment systems, reporting modules, and analytics features (to be considered in future development).

E. Primary Use Cases

In the SIMAA UGM system, organizers create and manage events by specifying event details such as title, venue, schedule, and capacity. Students can browse available events, register to participate, and cancel their registrations when needed. Organizers can also monitor registration data to ensure effective event management and venue utilization.

II. DATABASE DESIGN



A. Entity Explanation

Entities:

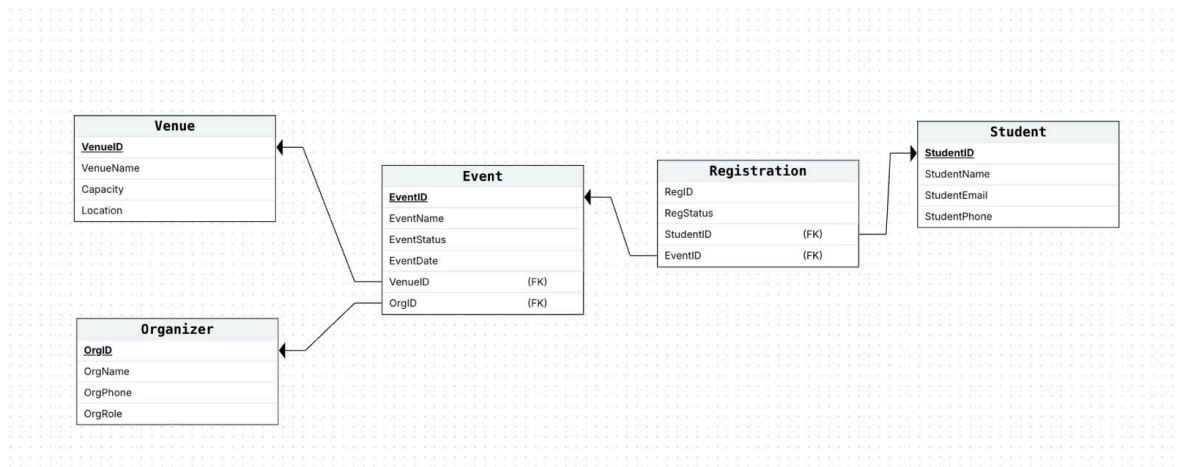
- Student
 - Attributes: StudentID (primary key), StudentName, StudentEmail, StudentPhone
- Event
 - Attributes: EventID (primary key), EventName, EventStatus, Category, EventDate
- Organizer
 - Attributes: OrgID (primary key), OrgName, OrgPhone, OrgRole
- Venue
 - Attributes: VenueID (primary key), VenueName, Location, Capacity
- Registration

- Attributes: RegID (primary key), RegDate, RegStatus

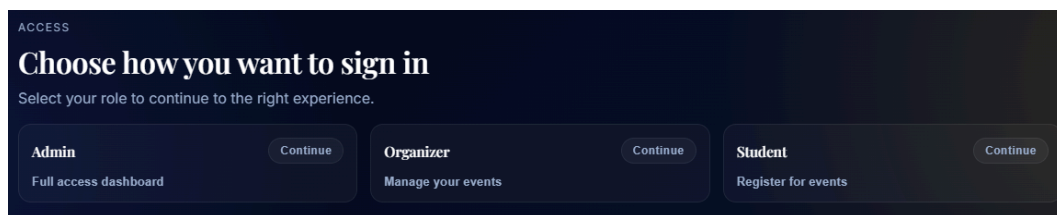
Relationship:

Relationship	Entities Involved	Cardinality	Explanation
RegisterFor	Event ↔ Student	Many-to-Many (M:N)	One student can register for multiple events One event can have multiple student registrations
Creates	Organizer ↔ Event	One-to-Many (1:N)	One organizer can create multiple events One event is created by one organizer
Has	Event ↔ Registration	One-to-Many (1:N)	One event can have multiple registrations Each registration is for exactly one event
HeldAt	Event ↔ Venue	Many-to-One (N:1)	One event is held at one venue One venue can host multiple events (at different times)

B. Logical Schema



III. SYSTEM OVERVIEW



SIMAA UGM allows different users to interact with the system according to their role

Role	Description
Admin	Full system control
Organizer	Event management
Student (User)	Event registration

Each role accesses the same system but with different permissions, ensuring data security and clarity of responsibility.

IV. USER ROLES AND ACCESS CONTROL

1. Admin

The Admin is responsible for managing the entire system.

Admin capabilities :

- 1) Manage students (CRUD)
- 2) Manage organizers (CRUD)
- 3) Manage venues (CRUD)
- 4) Manage events (CRUD)
- 5) Manage registrations
- 6) View all system data

The Admin acts as a system supervisor, ensuring data consistency and correctness.

2. Organizer

SIMAA UGM
Campus Events

Organizer Logout

ORGANIZER

Welcome, OmahTI

Manage your organization and events. Other data is view-only.

Logout

Your organization

OmahTI 081237 BSO

Save organization Editable

Your events

1 editable Create Event

Edit event

OmahTI Learning Center Published Workshop 02/03/2026

Auditorium

Save event Cancel

OmahTI Learning Center - Workshop
Auditorium • Published
Date: 2026-03-02

Edit Editable

Venues (view only)

2 venues

Auditorium
Location: FMIPA UGM
Capacity: 50

DIKE Classroom
Location: FMIPA UGM
Capacity: 2

Registrations (view only)

1 records

OmahTI Learning Center
Student: Johannes • Date: 2025-12-21
Registered

The Organizer represents departments, committees, or event hosts.

Organizer capabilities:

- 1) Create and update events
- 2) Assign venues to events
- 3) View registered students for their events
- 4) Update event status (Draft, Published, Closed)

Organizers cannot modify student or venue data, enforcing separation of duties.

3. Student (User)

The screenshot shows a web application interface for 'SIMAA UGM Campus Events'. At the top, there's a header with the logo and name, and two buttons: 'Student' and 'Logout'. Below the header, the user is logged in as 'Johannes'. The main content area is divided into several sections: 1. 'Your profile' section with fields for name (Johannes), email (john@mail.com), and a unique ID (1234). There's a 'Save profile' button and an 'Editable' status indicator. 2. 'Register for an event' section with a dropdown for 'Select event', a date field set to '22/12/2025', a 'Registered' status dropdown, and a 'Submit registration' button. 3. 'Your registrations' section showing one registration for 'OmahTI Learning Center' on '2026-03-02' at 'Auditorium', with a 'Registered' status and '1 records' count. 4. 'Events (view only)' section showing one event 'OmahTI Learning Center - Workshop' on '2026-03-02' at 'Auditorium', with a 'View only' status and '1 available' count.

The Student is the end user of the system.

Student capabilities:

- 1) View available events
- 2) Register for events
- 3) Cancel registration
- 4) View registration status

Students cannot modify event or system data, ensuring system integrity.

V. SYSTEM ARCHITECTURE

- Presentation Layer : Next.js (React)
- Application Layer : Node.js

- Database : MySQL relational database

VI. USER INTERFACE

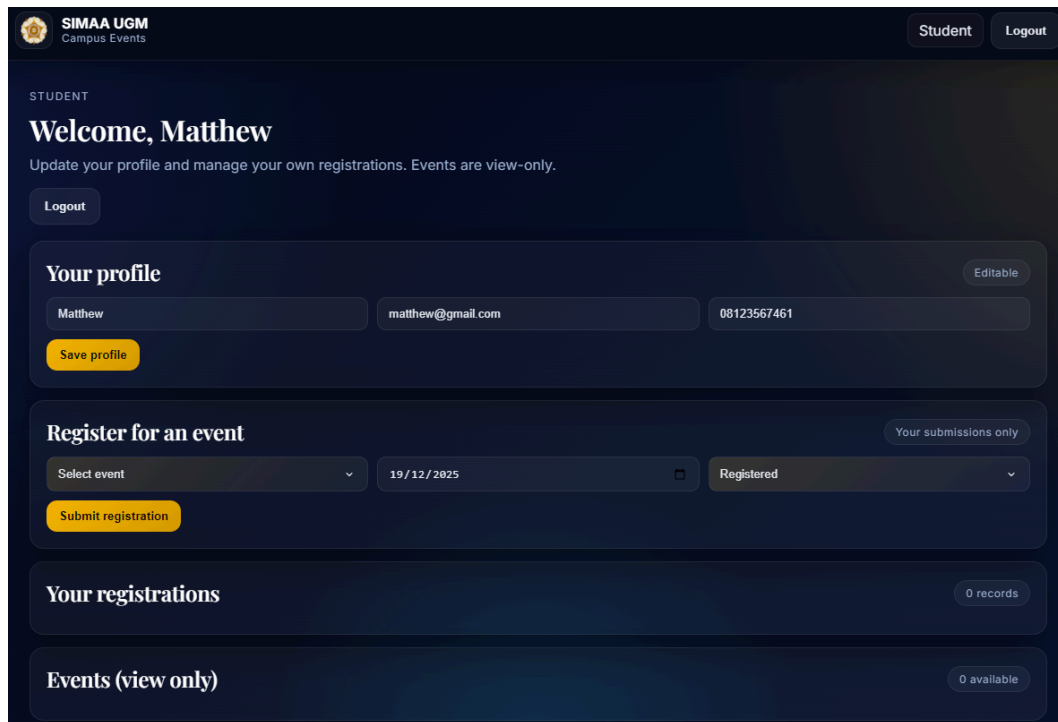
- Admin View

The screenshot shows the 'Student registrations' page in the SIMAA UGM system. The header includes the SIMAA UGM logo and navigation buttons for Registration, Events, Venues, Students, Organizers, and Logout. The page title is 'Student registrations' with a subtitle 'Keep attendance capped to venue capacity and maintain clean records.' Below this is an 'Add registration' form with dropdowns for 'Select Event', 'Select Student', a date field 'dd/mm/yyyy', and a 'Registered' status dropdown. A yellow 'Add' button is at the bottom left of the form. Below the form is a 'Registration list' section showing '0 total' records.

- Organizer View

The screenshot shows the 'Organizer View' for 'OmahTI' in the SIMAA UGM system. The header includes the SIMAA UGM logo and navigation buttons for Organizer and Logout. The page title is 'Welcome, OmahTI' with a subtitle 'Manage your organization and events. Other data is view-only.' Below this is a 'Logout' button. The main content area is divided into four sections: 'Your organization' (with fields for 'OmahTI', '081237', and 'yes', and a yellow 'Save organization' button), 'Your events' (showing '0 editable'), 'Venues (view only)' (showing '2 venues' with details for 'Auditorium' and 'DIKE Classroom'), and 'Registrations (view only)' (showing '0 records').

- User view



SIMAA UGM
Campus Events

Student Logout

STUDENT

Welcome, Matthew

Update your profile and manage your own registrations. Events are view-only.

Logout

Your profile

Editable

Matthew matthew@gmail.com 08123567461

Save profile

Register for an event

Your submissions only

Select event 19 / 12 / 2025 Registered

Submit registration

Your registrations

0 records

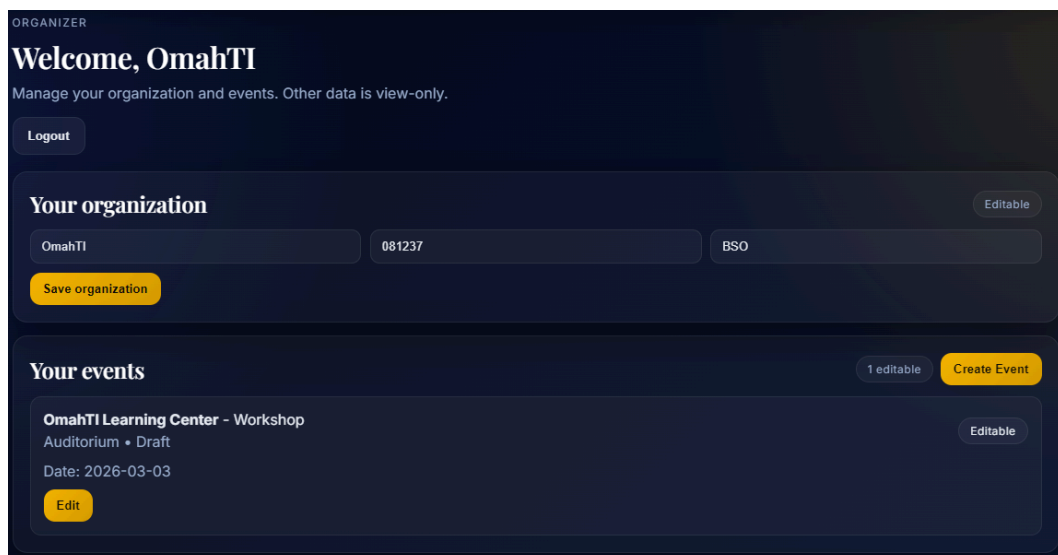
Events (view only)

0 available

VII. FUNCTIONAL DOCUMENTATION

- Event Management Flow

1) Organizer creates an event



ORGANIZER

Welcome, OmahTI

Manage your organization and events. Other data is view-only.

Logout

Your organization

Editable

OmahTI 081237 BSO

Save organization

Your events

1 editable Create Event

OmahTI Learning Center - Workshop

Auditorium • Draft

Date: 2026-03-03

Edit

2) Admin reviews and publishes event

EVENTS

Events & schedules

Plan, publish, and track campus happenings.

Add event

Select Organizer

Select Venue

Event Name

Draft

Category

dd/mm/yyyy

Add

Event list

1 total

OmahTI Learning Center - Workshop

OmahTI • Auditorium

Date: 2026-03-02

EditDelete

Published

3) Students view available events

STUDENT

Welcome, Johannes

Update your profile and manage your own registrations. Events are view-only.

Logout

Your profile

Editable

Johannes

john@mail.com

1234

Save profile

Register for an event

Your submissions only

Select event

22/12/2025

Registered

Submit registration

Your registrations

0 records

Events (view only)

1 available

OmahTI Learning Center - Workshop

Published • Auditorium

Date: 2026-03-02

View only

4) Students register

STUDENT

Welcome, Johannes

Update your profile and manage your own registrations. Events are view-only.

Logout

Your profile

Johannes

john@mail.com

1234

Save profile

Register for an event

Select event

22/12/2025

Registered

Submit registration

Your registrations

OmahTI Learning Center

2026-03-02 • Auditorium

Registered on: 2025-12-21

Registered

5) Organizer monitors registrations

SIMAA UGM
Campus Events

Organizer

Logout

OmahTI

081237

BSO

Save organization

Your events

OmahTI Learning Center - Workshop

Auditorium • Published

Date: 2026-03-02

Edit

Venues (view only)

Auditorium

Location: FMIPA UGM

Capacity: 50

DIKE Classroom

Location: FMIPA UGM

Capacity: 2

Registrations (view only)

OmahTI Learning Center

Student: Johannes • Date: 2025-12-21

Registered

- Registration Workflow

- Student selects an event

STUDENT

Welcome, Johannes

Update your profile and manage your own registrations. Events are view-only.

[Logout](#)

Your profile

[Editable](#)

Johannes john@mail.com 1234

[Save profile](#)

Register for an event

[Your submissions only](#)

Select event 22/12/2025 Registered

[Submit registration](#)

Your registrations

[1 records](#)

OmahTI Learning Center [Registered](#)

2026-03-02 • Auditorium
Registered on: 2025-12-21

2) System validates event status

 **SIMAA UGM**
Campus Events

[Registration](#) [Events](#) [Venues](#) [Students](#) [Organizers](#) [Logout](#)

EVENTS

Events & schedules

Plan, publish, and track campus happenings.

Add event

[Draft](#)

Select Organizer Select Venue Event Name Draft

Category dd/mm/yyyy

[Add](#)

Event list

[1 total](#)

OmahTI Learning Center - Workshop [Published](#)

OmahTI • Auditorium
Date: 2026-03-02

[Edit](#) [Delete](#)

3) Registration record is created

**SIMAA UGM**
Campus Events

[Registration](#)[Events](#)[Venues](#)[Students](#)[Organizers](#)[Logout](#)

REGISTRATION

Student registrations

Keep attendance capped to venue capacity and maintain clean records.

Add registration

Select Event

Select Student

dd/mm/yyyy

Registered

Add

Registration list

1 total

OmahTI Learning Center - Johannes

Registered

Date: 2025-12-21

EditDelete

4) Registration status is stored and tracked